

Use Case Descriptions

Chapter-07

Diagram 1 – Customer Booking Flow

UC-001: Search Available Buses

Use Case ID	UC-001
Use Case Name	Search Available Buses
Actor(s)	Customer
Description	Customer searches for available buses by selecting origin, destination, and travel date.
Preconditions	Customer is logged in. Bus routes and schedules have been configured by Admin.
Postconditions – Success	Customer views available buses and proceeds to booking.
Main Flow	1. Customer navigates to the bus search section. 2. Customer selects origin, destination, and travel date. 3. System queries available buses. 4. System displays list of available buses with seat availability. 5. Customer selects a preferred bus.
«include»	—
«extend»	—

UC-002: Book Bus Ticket

Use Case ID	UC-002
Use Case Name	Book Bus Ticket
Actor(s)	Customer, System
Description	Customer selects and confirms a seat reservation on an available bus route.
Preconditions	Customer is logged in. Customer has searched and selected an available bus. Seats are available.
Postconditions – Success	Seat(s) reserved. Booking confirmation sent. Seat availability updated.

Postconditions – Failure	Booking not confirmed. Seat reservation released after timeout.
Main Flow	1. Customer selects one or more available seats. 2. System temporarily reserves seat(s) for 15 minutes. 3. Customer reviews and confirms booking. 4. System processes and confirms the booking. 5. Confirmation sent via email or SMS.
Alternative Flow 1	At step 1: If no seats are available, system displays a 'Sold Out' message and suggests alternative times or routes.
Alternative Flow 2	At step 3: If customer does not confirm within 15 minutes, system releases the temporary reservation.
Business Rules	BR-01: Seat reservations expire after 15 minutes. BR-02: Booking confirmation must be sent within 2 minutes of confirmation.
«include»	Reserve seat (15 min)
«extend»	Cancel booking

UC-003: Book Airport Transfer

Use Case ID	UC-003
Use Case Name	Book Airport Transfer
Actor(s)	Customer, Admin
Description	Customer books an airport pickup or drop-off by providing travel details.
Preconditions	Customer is logged in. Airport transfer services are available.
Postconditions – Success	Transfer booked and confirmed. Customer receives notification.
Main Flow	1. Customer navigates to airport transfer section. 2. Customer provides date, time, location, and flight details. 3. System displays available vehicles and estimated pricing. 4. Customer selects vehicle and confirms booking. 5. System generates confirmation and notifies customer.
«include»	Calculate pricing
«extend»	Cancel booking

UC-004: Customise Tour Package

Use Case ID	UC-004
Use Case Name	Customise Tour Package
Actor(s)	Customer, System
Description	Customer creates a personalised Sri Lankan tour package by selecting tourist destinations.
Preconditions	Customer is logged in. Tourist destinations and pricing configured by Admin.
Postconditions – Success	Tour package created and confirmed. Booking visible in customer history.
Main Flow	1. Customer navigates to tour package section. 2. Customer selects destinations from available list. 3. Customer selects vehicle type and specifies duration. 4. System calculates total package price. 5. Customer reviews, saves, and submits the tour booking. 6. System generates confirmation and notifies customer.
«include»	Calculate tour price
«extend»	Cancel booking

UC-005: Cancel Booking

Use Case ID	UC-005
Use Case Name	Cancel Booking
Actor(s)	Customer, System
Description	Customer cancels an existing bus, airport transfer, or tour booking within the permitted cancellation window.
Preconditions	Customer is logged in. Customer has an active confirmed booking. Cancellation is within the permitted time window.
Postconditions – Success	Booking status updated to Cancelled. Notifications sent. Availability restored.
Postconditions – Failure	Cancellation not permitted. Customer notified of cancellation window violation.
Main Flow	1. Customer navigates to booking history. 2. Customer selects the booking to cancel. 3. System validates cancellation is within permitted window. 4. Customer confirms cancellation. 5. System updates booking status to Cancelled. 6. System sends cancellation confirmation to customer and Admin.

	7. System restores seat or vehicle availability.
Alternative Flow 1	At step 3: If cancellation window has expired, system displays an error message and denies the cancellation request.
Business Rules	BR-03: Cancellations must be requested at least 2 hours before scheduled departure.
«include»	Send cancellation notice
«extend»	Triggered from: Book bus ticket, Book airport transfer, Customise tour package

Diagram 2 – Authentication & Account Access

UC-006: Register / Login

Use Case ID	UC-006
Use Case Name	Register / Login
Actor(s)	Customer
Description	A new customer registers an account or an existing customer authenticates to access the platform.
Preconditions	User has access to the A9 Express platform. User has a valid email address.
Postconditions – Success	Customer account created or authenticated. Session initiated. Customer redirected to dashboard.
Postconditions – Failure	Account not created or login denied. Error message displayed.
Main Flow	1. New customer navigates to registration page and provides name, email, phone, and password. 2. System validates details and creates the account. 3. Existing customer enters valid email and password. 4. System authenticates the user. 5. System initiates session and redirects to dashboard.
Alternative Flow 1	At step 1 (Registration): If email is already registered, system notifies user and prompts login instead.
Alternative Flow 2	At step 3 (Login): If credentials are invalid after 3 attempts, account is temporarily locked for 15 minutes.
Business Rules	BR-06: Passwords must be at least 8 characters with one uppercase letter and one number. BR-07: Account lockout after 3 failed login attempts.
«include»	Validate credentials, Create user account, Initiate session, Redirect to dashboard
«extend»	View booking history

UC-007: Validate Credentials

Use Case ID	UC-007
Use Case Name	Validate Credentials
Actor(s)	System
Description	System validates the email and password provided by the customer during login.
Preconditions	Customer has submitted login credentials.
Postconditions – Success	Credentials validated; authentication proceeds.
Postconditions – Failure	Invalid credentials; error message returned to customer.
Main Flow	1. System receives email and password input. 2. System checks credentials against stored records. 3. If valid, authentication proceeds. 4. If invalid, system returns an error message.
«include»	—
«extend»	—

UC-008: Initiate Session

Use Case ID	UC-008
Use Case Name	Initiate Session
Actor(s)	System
Description	System creates an authenticated session for the customer after successful login.
Preconditions	Credentials have been successfully validated.
Postconditions – Success	Active session created. Customer has access to the platform.
Main Flow	1. System generates a session token for the authenticated user. 2. Session is stored and associated with the customer account. 3. Customer is granted access to protected platform features.
«include»	—
«extend»	—

UC-009: View Booking History

Use Case ID	UC-009
Use Case Name	View Booking History
Actor(s)	Customer
Description	Customer views all current and past bookings from their dashboard after login.
Preconditions	Customer is logged in.
Postconditions – Success	Customer views complete booking history. Booking details displayed accurately.
Main Flow	1. Customer navigates to booking history section on dashboard. 2. System retrieves all current and past bookings. 3. Customer views booking details including service type, date, time, and payment status. 4. Customer selects a specific booking to view full details.
«include»	—
«extend»	Extends Register / Login

Diagram 3 – Admin Route & Vehicle Management

UC-010: Manage Routes & Schedules

Use Case ID	UC-010
Use Case Name	Manage Routes & Schedules
Actor(s)	Admin
Description	Admin creates, updates, and deactivates bus routes and configures schedules.
Preconditions	Admin is logged in with appropriate permissions.
Postconditions – Success	New route and schedule are active and bookable. Seat availability initialised.
Main Flow	1. Admin navigates to route management section. 2. Admin creates a new route by defining origin, destination, and stops. 3. Admin configures schedules including departure times and days of operation. 4. Admin sets ticket pricing per route. 5. System saves and makes the route available for customer booking.
«include»	Configure schedule, Set seat capacity
«extend»	—

UC-011: Manage Vehicles & Drivers

Use Case ID	UC-011
Use Case Name	Manage Vehicles & Drivers
Actor(s)	Admin
Description	Admin manages the fleet of vehicles and driver profiles on the platform.
Preconditions	Admin is logged in with appropriate permissions.
Postconditions – Success	Vehicle and driver profiles saved and available for assignment.
Main Flow	1. Admin navigates to vehicle management section. 2. Admin adds a new vehicle with type, capacity, and registration details. 3. Admin navigates to driver management section. 4. Admin creates or updates a driver profile with personal details and licence information. 5. Admin sets driver status as active or inactive. 6. System saves vehicle and driver information.
«include»	Register vehicle, Create driver profile
«extend»	Assign driver to booking

UC-012: Assign Driver to Booking

Use Case ID	UC-012
Use Case Name	Assign Driver to Booking
Actor(s)	Admin, Driver
Description	Admin assigns a driver and vehicle to a confirmed airport transfer or tour booking.
Preconditions	Admin is logged in. A confirmed booking exists requiring a driver. An available driver and vehicle exist.
Postconditions – Success	Driver and vehicle assigned to booking. Vehicle availability updated. Driver notified.
Main Flow	1. Admin views confirmed bookings requiring driver assignment. 2. Admin selects a booking to assign. 3. Admin selects an available driver and vehicle. 4. System assigns driver and vehicle to the booking. 5. System updates vehicle availability. 6. Driver is notified of the assignment.
«include»	Notify driver assigned
«extend»	Extends Manage vehicles & drivers

Diagram 4 – Notification & System Events

UC-013: Send Booking Notifications

Use Case ID	UC-013
Use Case Name	Send Booking Notifications
Actor(s)	System
Description	System automatically sends real-time notifications to customers and Admin on booking events.
Preconditions	A booking, cancellation, or assignment event has occurred. Customer contact details are available.
Postconditions – Success	Customer receives booking or cancellation confirmation. Admin notified of relevant events.
Main Flow	1. System detects a booking confirmation, cancellation, or driver assignment event. 2. System retrieves customer contact details. 3. System generates the appropriate notification message. 4. System dispatches notification via email or SMS. 5. System notifies Admin of relevant booking events.
«include»	Retrieve contact details, Dispatch via email/SMS, Notify Admin of event
«extend»	Booking confirmation event, Cancellation event, Driver assignment event

UC-014: Booking Confirmation Event

Use Case ID	UC-014
Use Case Name	Booking Confirmation Event
Actor(s)	System
Description	System triggers a notification when a new booking is successfully confirmed.
Preconditions	A booking has been successfully created and confirmed.
Postconditions – Success	Customer notified of confirmed booking.
Main Flow	1. System detects booking status set to Confirmed. 2. System triggers Send booking notifications use case. 3. Customer receives booking confirmation with details.
«include»	—
«extend»	Extends Send booking notifications

UC-015: Cancellation Event

Use Case ID	UC-015
Use Case Name	Cancellation Event
Actor(s)	System
Description	System triggers a notification when a booking is cancelled by the customer.
Preconditions	A booking has been successfully cancelled.
Postconditions – Success	Customer and Admin notified of cancellation.
Main Flow	1. System detects booking status updated to Cancelled. 2. System triggers Send booking notifications use case. 3. Customer and Admin receive cancellation confirmation.
«include»	—
«extend»	Extends Send booking notifications

UC-016: Driver Assignment Event

Use Case ID	UC-016
Use Case Name	Driver Assignment Event
Actor(s)	System
Description	System triggers a notification when a driver is assigned to a confirmed booking.
Preconditions	A driver has been successfully assigned to a booking by Admin.
Postconditions – Success	Driver notified of booking assignment.
Main Flow	1. System detects driver assignment to a confirmed booking. 2. System triggers Send booking notifications use case. 3. Driver receives assignment notification with booking details.
«include»	—
«extend»	Extends Send booking notifications

Diagram 5 – Admin Reporting & Analytics

UC-017: Generate Reports

Use Case ID	UC-017
Use Case Name	Generate Reports
Actor(s)	Admin
Description	Admin generates operational reports and views analytics on bookings and revenue.
Preconditions	Admin is logged in. Booking and operational data exists in the system.
Postconditions – Success	Report generated and displayed to Admin.
Main Flow	1. Admin navigates to reporting and analytics section. 2. Admin selects report type (bookings, revenue, vehicle utilisation). 3. Admin specifies date range and filters. 4. System generates and displays the report. 5. Admin views analytics dashboard with booking trends and revenue summaries.
«include»	Select report type, Apply date range filter, Display analytics dashboard
«extend»	Access audit logs, Export report

UC-018: Access Audit Logs

Use Case ID	UC-018
Use Case Name	Access Audit Logs
Actor(s)	Admin
Description	Admin optionally accesses detailed audit logs for deeper review of system activity.
Preconditions	Admin has generated a report. Admin requires detailed activity tracking.
Postconditions – Success	Audit logs displayed and accessible for review.
Main Flow	1. Admin selects the audit log option from the reporting section. 2. System retrieves audit log entries. 3. Admin applies filter by vehicle or route. 4. System displays filtered audit log results.
«include»	Filter by vehicle / route
«extend»	Extends Generate reports

UC-019: Export Report

Use Case ID	UC-019
Use Case Name	Export Report
Actor(s)	Admin
Description	Admin optionally exports a generated report for external use or record keeping.
Preconditions	Admin has generated a report.
Postconditions – Success	Report exported and downloaded by Admin.
Main Flow	1. Admin selects the export option after viewing a report. 2. Admin chooses the preferred export format. 3. System generates the export file. 4. Admin downloads the file.
«include»	—
«extend»	Extends Generate reports